Water Pollution

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What is Water Quality?

- Water quality refers to the physical, chemical, and bacteriological properties of water, which are measured based on specific usage requirements and quality concerns. It is determined by a set of 16 basic parameters that are essential for ensuring the quality of water for various purposes such as domestic, agriculture, industrial, and fisheries.

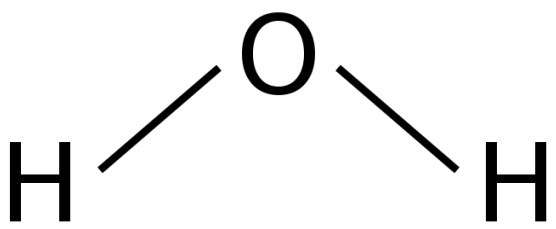
Source: 1. Water Quality, Sciencedirect, https://www.sciencedirect.com/topics/earth-and-planetary-sciences/water-quality
2. Image: Tektelic, https://tektelic.com/expertise/smart-water-quality-monitoring-why-is-it-better-than-traditional-methods/

Chemistry of Solution

- An ATOM is the smallest unit of each element .
- Atoms are building blocks from which MOLECULES of elements and compounds are constructed.
- Two hydrogen atoms combine to form a molecular of hydrogen gas.
- Molecular mass of Hydrogen=1 g/mol
- Molecular mass of Oxygen= 16 g/mol

Molecular mass of water = (2x1)+16=18 g/mol

$$H + H \longrightarrow H_2$$



Water Molecule

$$2H_2+O_2 \rightarrow 2H_2O$$

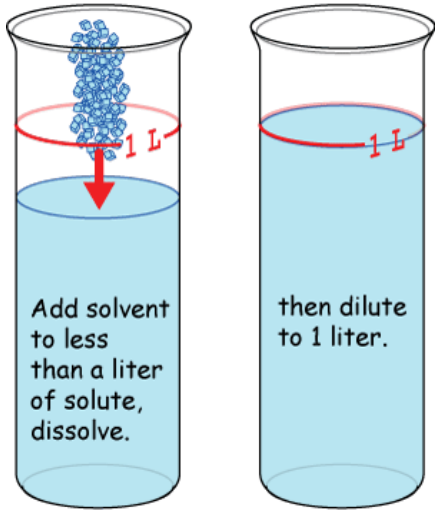
Source: 1. Howard S. Peavy et al, Environmental Engineering, McGraw Hill Education (2018). Image source: Datastream, https://datastream.org/en-ca/guidebook/total-suspended-solids-tss-and-total-dissolved-solids-tds

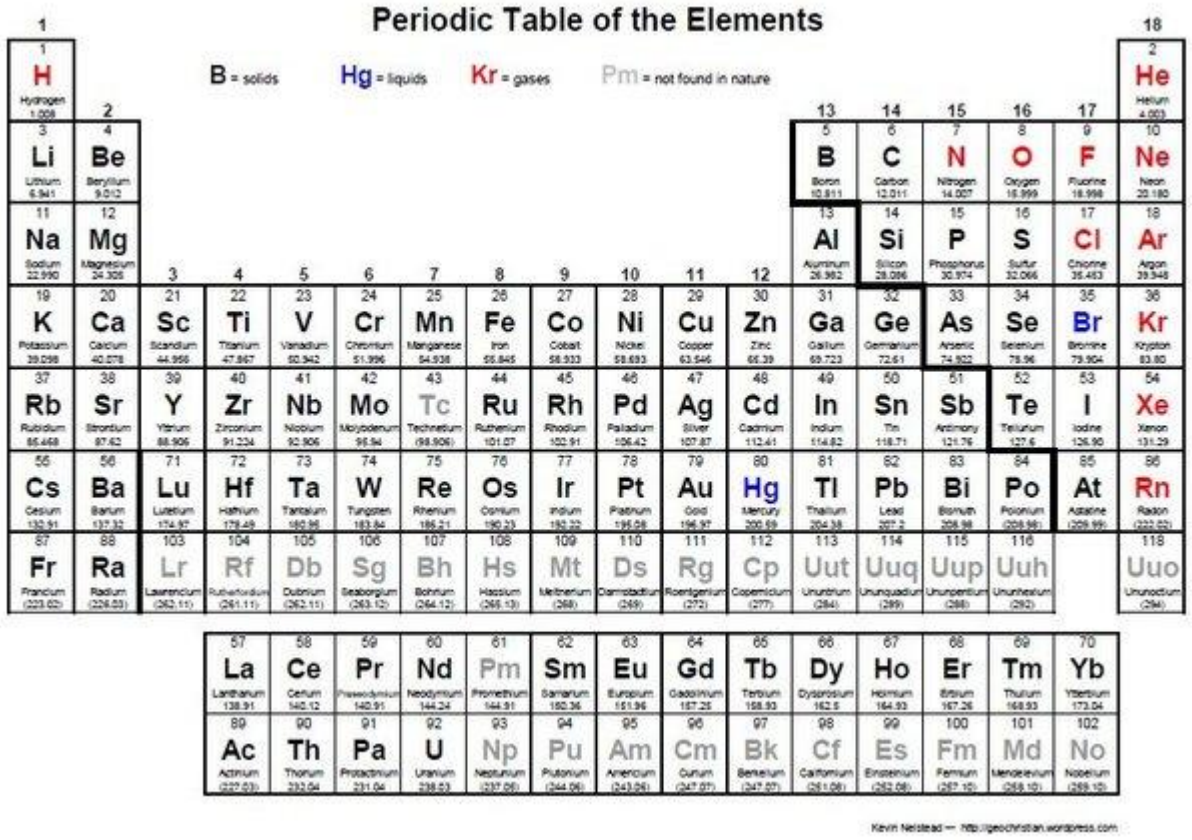
Molarity

- Molarity is a unit of concentration in chemistry that measures the number of moles of solute dissolved in a liter of solution.

- Unit- mol/L

$$M = \frac{n\,(\text{moles of solute})}{V\,(\text{volume of solution})}$$





Source: 1. Molarity, Sciencedirect, https://www.sciencedirect.com/topics/chemistry/molarity

Numerical

Question No. 1: A solution is prepared using 20 grams of Sodium Chloride in 250 ml of water, what is the molarity of the solution?

Answer: NaCl = 23+35.5 = 58.5 grams/mol

$$\text{Molarity}=\frac{20/58.5}{250/1000} = 1.4\ mol/L$$

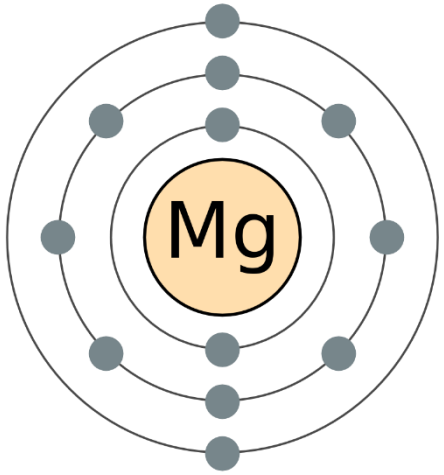
Question No. 2: A solution is prepared using 15 grams of Potassium hydroxide in 300 ml of water, what is the molarity of the solution?

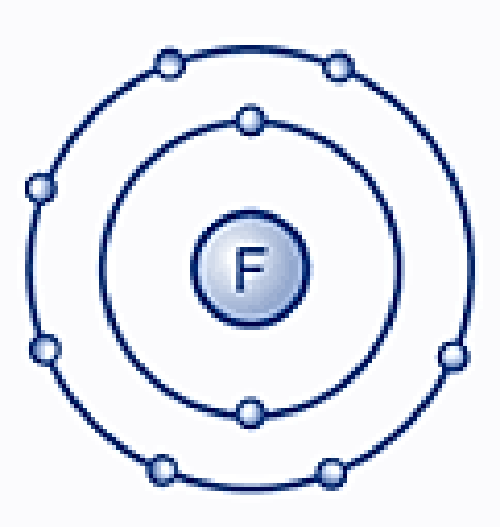
Source: 1. Howard S. Peavy et al, Environmental Engineering, McGraw Hill Education (2018).

Normality

- Electrons in an atom are arranged in different orbitals (shells) represented as K, L, M, N, and so on.
- The electrons present in the outermost shell/orbit of an atom are called **valence electrons**.
- The valence electrons take part in any chemical reaction because the outermost orbit usually contains more energy than the electrons present in other orbits.

- **Equivalent** is the amount of substance that reacts with one mole of another substance in a given chemical reaction.





Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Valency, https://www.toppr.com/guides/chemistry/structure-of-atom/valency/, Accessed on: 14 January, 2025

Normality

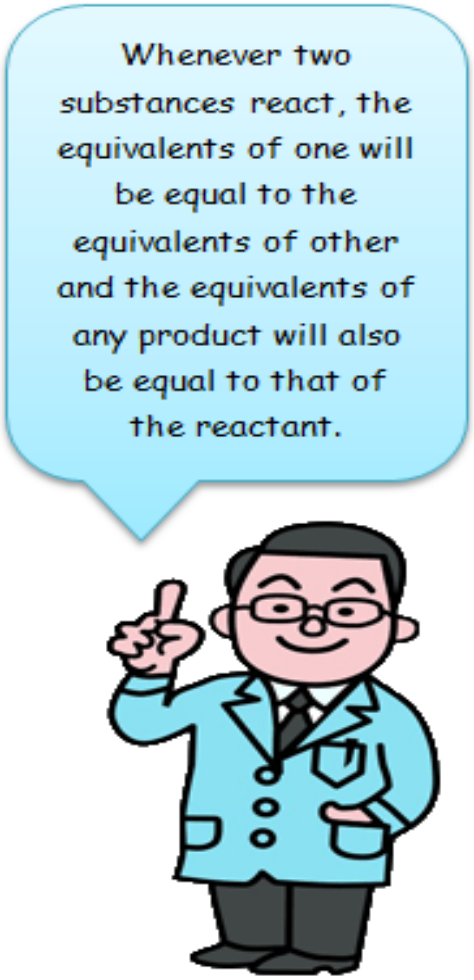
- The equivalence of an ion is the same as the absolute value of its valence.
- An equivalent of an element or radical is its gram molecular mass divided by its equivalence.

$$Equivalent = \frac{Molecular\ weight}{Equivalence}\ g/equiv$$

- A milliequivalent is the molecular mass expressed in milligrams divided by the equivalence.

- ***Normality*** *is the number of equivalents of a solute divided by the liters of a solution.*

$$Normality = \frac{Equivalent\ of\ solute}{volume\ of\ solution}\ N$$



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Law of equivalence, https://www.askiitians.com/iit-jee-chemistry/physical-chemistry/stoichiometry-and-redox-reactions/law-of-equivalents.aspx, Accessed on: 14 January, 2025

Numerical

Question No. 3: How many grams of calcium will be required to combine with 90 g of carbonate to form calcium carbonate?

Answer:

1. Carbonate (CO_3^{2-}) is a radical composed of carbon and oxygen. In this particular combination, carbon has an atomic mass of 12 and a valence of +4, while oxygen has an atomic mass of 16 and a valence of 2. Therefore, the radical has a total valence of -2 and an equivalence of 2.

$$\text{One equivalent of carbonate} = \frac{12+3(16)}{2} = 30 \text{ g/equiv.}$$

2. The calcium ion has an atomic mass of 40 and a valence of +2. Therefore, one equivalent of calcium is 40÷2 = 20 g/equiv.
3. The number of equivalents of calcium must equal the number of equivalents of carbonate, therefore

$$\frac{90\ g}{30 \text{ g/equiv}} = 3 \text{ equiv of carbonate.}$$

Therefore, 3 equiv x 20 g/equiv = 60 g of calcium, and that amount will be required to react with 90 g of carbonate

Source: 1. Howard S. Peavy et al, Environmental Engineering, McGraw Hill Education (2018).

Numerical

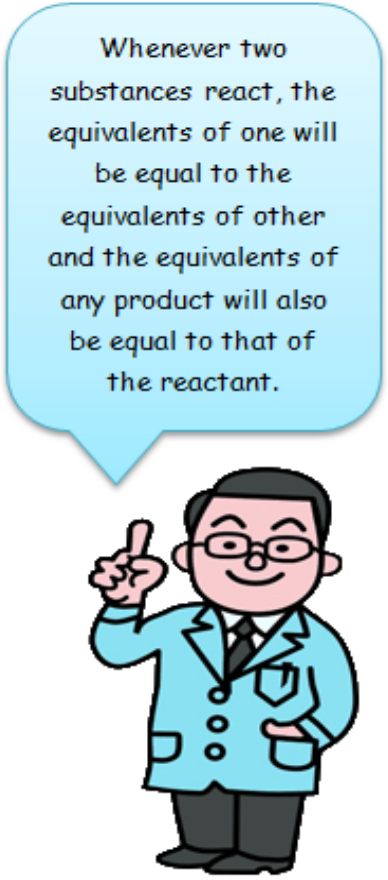
Question No. 4: what is the equivalent calcium carbonate concentration of (a) 117 mg/L of NaCl (b) 2×10^{-3} mg/L of NaCl?

Answer:

(a) One equivalent of calcium carbonate = $\frac{40+12+3(16)}{2}$ = 50 g/equiv.

One equivalent of Sodium Chloride= $\frac{23+35.5}{1}$ = 58.5 g/equiv

$$\frac{117\ mg/l}{58.5\ g/equiv} \times 50 \frac{g}{equiv} = 100 \frac{mg}{l}\ of\ NaCl\ as\ CaCO_3$$



The concentration of substance A can be expressed as an equivalent concentration of substance B using following expression:

$$\frac{\left(\frac{g}{l}\right)A}{\left(\frac{g}{equiv}\right)A} \times \left(\frac{g}{equiv}\right)B = \left(\frac{g}{l}\right)A\ expressed\ as\ B$$

(b) Answer: 100 mg/L

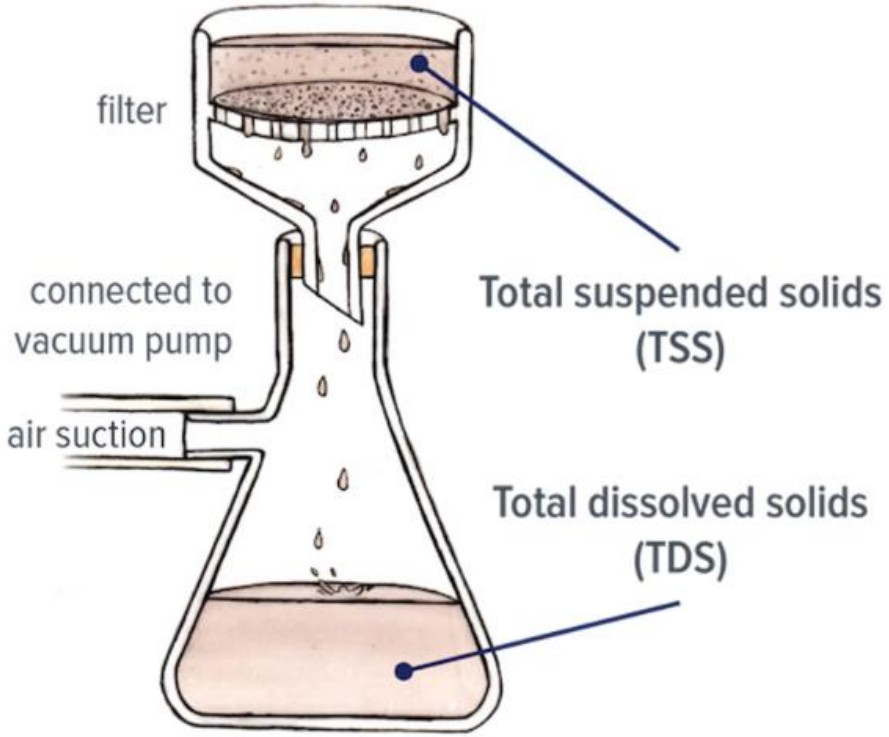
Source: 1. Howard S. Peavy et al, Environmental Engineering, McGraw Hill Education (2018).

Total Dissolved Solids (TDS)

- TDS is total concentration of dissolved substances in water.

- TDS is made up of inorganic salts, as well as a small amount of organic matter. Common inorganic salts that can be found in water include calcium, magnesium, potassium and sodium, which are all cations, and carbonates, nitrates, bicarbonates, chlorides and sulfates, which are all anions. Cations are positively charged ions and anions are negatively charged ions.

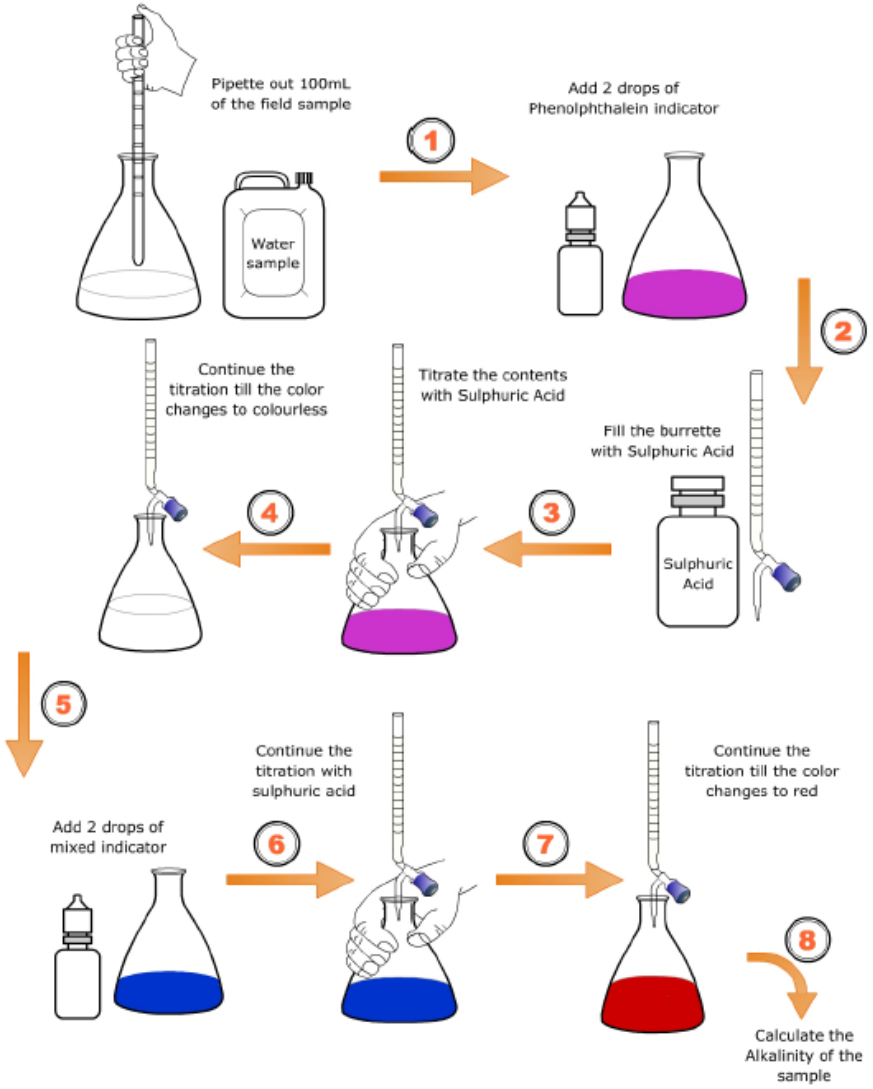
- As per IS 10500: 2012, the permissible limit of TDS in drinking water source shouldn't be more than 500 mg/L.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. APHA. (1992). Standard methods for the examination of water and wastewater. 18th ed., 3. Datastream, https://datastream.org/en-ca/guidebook/total-suspended-solids-tss-and-total-dissolved-solids-tds Accessed on: 14 January, 2025, 3. IS 10500:2012, Drinking Water — Specification

Alkalinity

- **Alkalinity is a measure of the capacity of water to neutralize acids.**
- Alkaline compounds in the water such as bicarbonates, carbonates, and hydroxides remove H+ ions and lower the acidity of the water (which means increased pH).
- They usually do this by combining with the H+ ions to make new compounds.
- Without this acid-neutralizing capacity, any acid added to a stream would cause an immediate change in the pH.
- Measuring alkalinity is important in determining a stream's ability to neutralize acidic pollution from rainfall or wastewater.
- It's one of the best measures of the sensitivity of the stream to acid inputs.
- **Total alkalinity is measured by measuring the amount of acid (e.g., sulfuric acid) needed to bring the sample to a pH of 4.2. At this pH all the alkaline compounds in the sample are "used up." The result is reported as milligrams per liter of calcium carbonate (mg/L $CaCO_3$).**



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. APHA. (1992). Standard methods for the examination of water and wastewater. 18th ed., 3. Datastream, https://datastream.org/en-ca/guidebook/total-suspended-solids-tss-and-total-dissolved-solids-tds Accessed on: 14 January, 2025

Alkalinity

$$H^+ + OH^- \rightleftharpoons H_2O$$

$$CO_3^{2-} + H^+ \rightleftharpoons HCO_3^-$$

$$HCO_3^- + H^+ \rightleftharpoons H_2CO_3$$

- If acid is added slowly to water and the pH is recorded for each addition. titration curve similar to that shown in Fig. is obtained. Of particular significance are the inflection points in the curve that occur at approximately pH 8.3 and pH 4.5
- The conversion of carbonate to bicarbonate is essentially complete at pH 8.3. However, because bicarbonate is also an alkalinity species, an equal amount of acid must be added to complete the neutralization. Thus, the neutralization of carbonate is only one-half complete at pH 8.3.
- Because the conversion hydroxide to water is virtually complete at pH 8.3 (see Fig.), all of the hydroxyl and one-half of the carbonate have been measured at pH 8.3.
- At pH 4.5 all of bicarbonate has been converted to carbonic acid, including bicarbonate resulting from the reaction of the acid and carbonate. Thus, the **amount of acid required to titrate a sample to pH 4.5 is equivalent to total alkalinity of the water.**

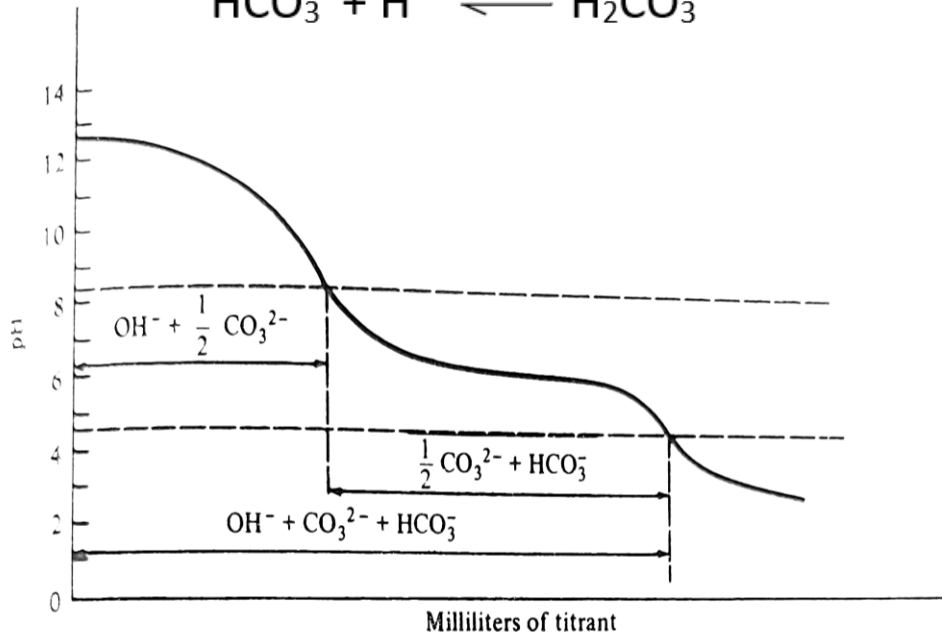


Figure 2-4 Alkalinity titration curve.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985),

Alkalinity

If the volume of acid needed to reach the 8.3 endpoint is known, the species of alkalinity can also be determined.

- Because all of the hydroxide and one-half of the carbonate have been neutralized at pH 8.3,
- the acid required to lower the pH from 8.3 to 4.5 must measure the other one-half of the carbonate, plus all of the original bicarbonate.
- If P is the amount of acid required to reach pH 8.3 and M is the total quantity of acid required to reach 4.5, the following generalizations concerning the forms of alkalinity can be made:

1. P = M all alkalinity is OH^-
2. P = M / 2 all alkalinity is CO_3^{2-}
3. P = 0 (i.e., initial pH is below 8.3), all alkalinity is HCO_3^-,
4. P < M / 2 predominant species are CO_3^{2-} and HCO_3^-,
5. P > M / 2 predominant species are OH^- and CO_3^{2-}

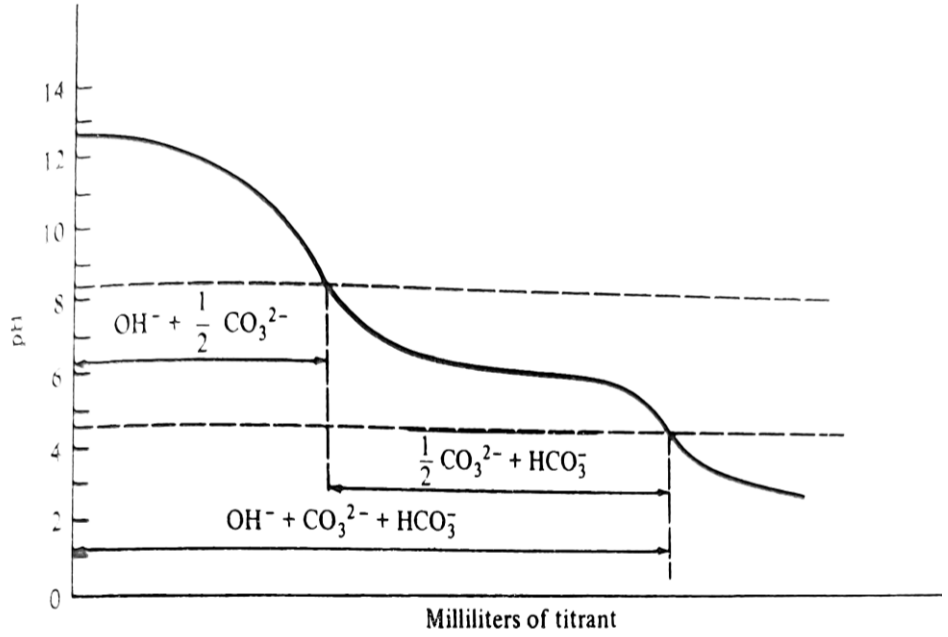
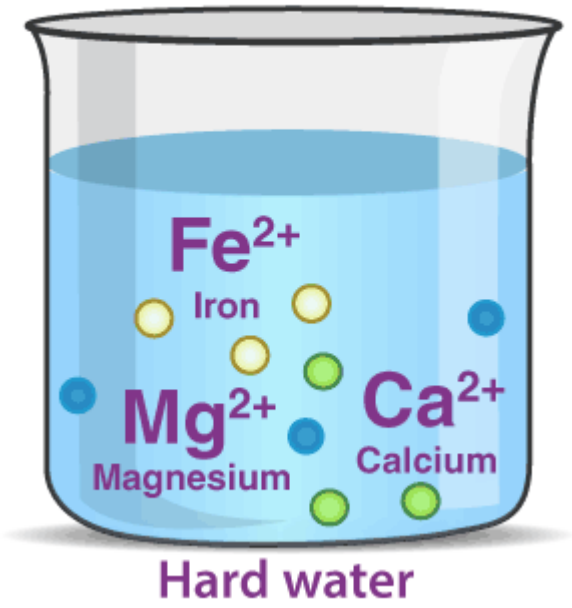


Figure 2-4 Alkalinity titration curve.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985),

Hardness

- *Hardness is defined as the concentration of multivalent metallic cations in solution.*
- At supersaturated conditions, the hardness cations will react with anions in the water to form a solid precipitate.
- Hardness is classified as carbonate hardness and non-carbonate hardness, depending upon the anion with which it associates.
- The hardness that is equivalent to the alkalinity is termed carbonate hardness, with any remaining hardness being called non-carbonate hardness.
- Carbonate hardness is sensitive to heat and precipitates readily at high temperatures.



$$Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 + H_2O + CO_2$$

$$Mg(HCO_3)_2 \xrightarrow{\Delta} MgCO_3 + H_2O + CO_2$$

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: Hardness, Byjus, https://byjus.com/jee/hardness-of-water-types-and-removal/

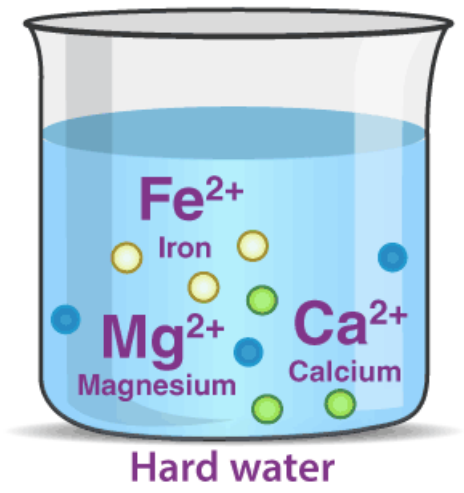
Hardness

Sources:

- The multivalent metallic ions most abundant in natural waters are calcium and magnesium.
- Others may include iron and manganese in their reduced states (Fe, Mn^{2+}), strontium (Sr^{2+}), and aluminum (Al^{3+}).
- The latter are usually found in much smaller quantities than calcium and magnesium, and *for all practical purposes, hardness may be represented by the sum of the calcium and magnesium ions.*

Impacts:

- Soap consumption by hard waters represents an economic loss to the water user. Sodium soaps react with multivalent metallic cations to form a precipitate, thereby losing their surfactant properties.
- Lathering does not occur until all of the hardness ions are precipitated, at which point the water has been "softened" by the soap. The precipitate formed by hard- ness and soap adheres to surfaces of tubs, sinks, and dishwashers and may stain clothing, dishes, and other items.
- Boiler scale, the result of the carbonate hardness precipitate may cause considerable economic loss through fouling of water heaters and hot-water pipes.



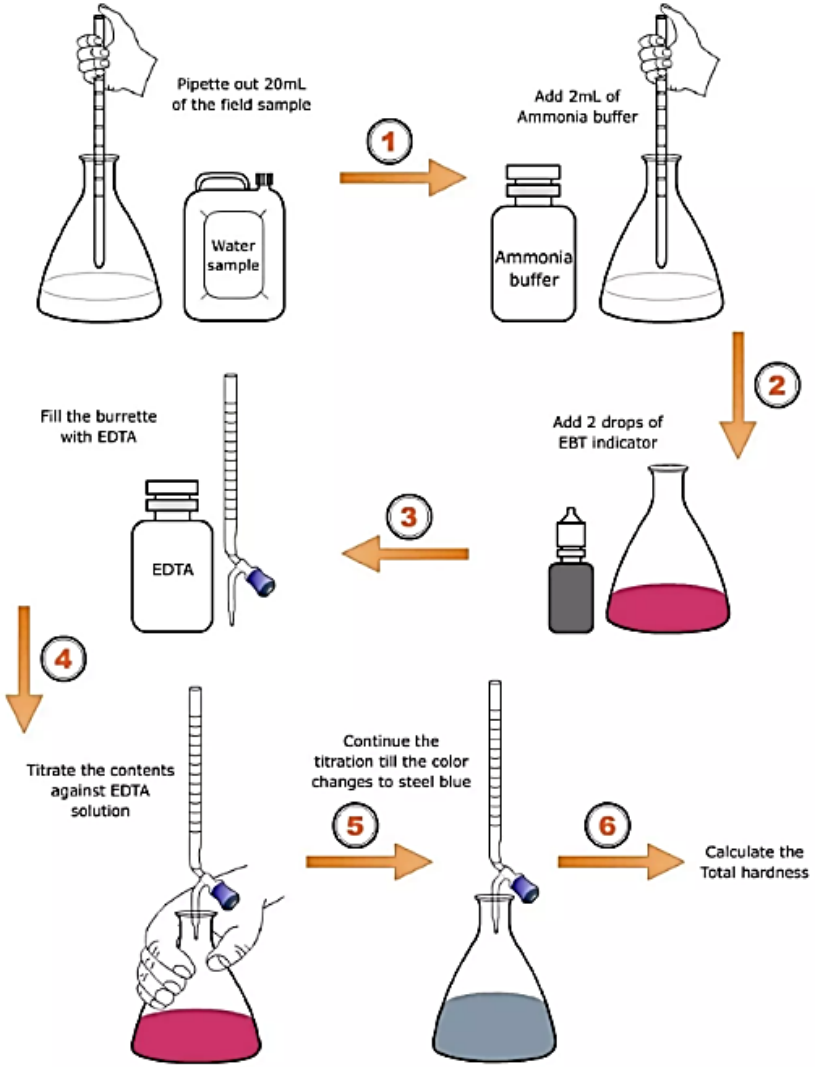
Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: Hardness, Byjus, https://byjus.com/jee/hardness-of-water-types-and-removal/

Hardness

Measurement:

- Hardness can be measured directly by titration with ethylenediamine tetraacetic acid (EDTA) using eriochrome black T (EBT) as an indicator.
- The EBT reacts with the divalent metallic cations, forming a complex that is red in color.
- The EDTA replaces the EBT in the complex, and when the replacement is complete the **solution changes from red to blue, If 0.01 M EDTA is used, 10 mL of titrant measures 1.0 mg of hardness as $CaCO_3$.**
- Although acceptability levels vary according to a consumer's acclimation to hardness, a generally accepted classification is as follows:

Soft	<50 mg/L. as $CaCO_3$
Moderately hard	50-150 mg/L as $CaCO_3$
Hard	150-300 mg/L as $CaCO_3$
Very hard	>300 mg/Las $CaCO_3$



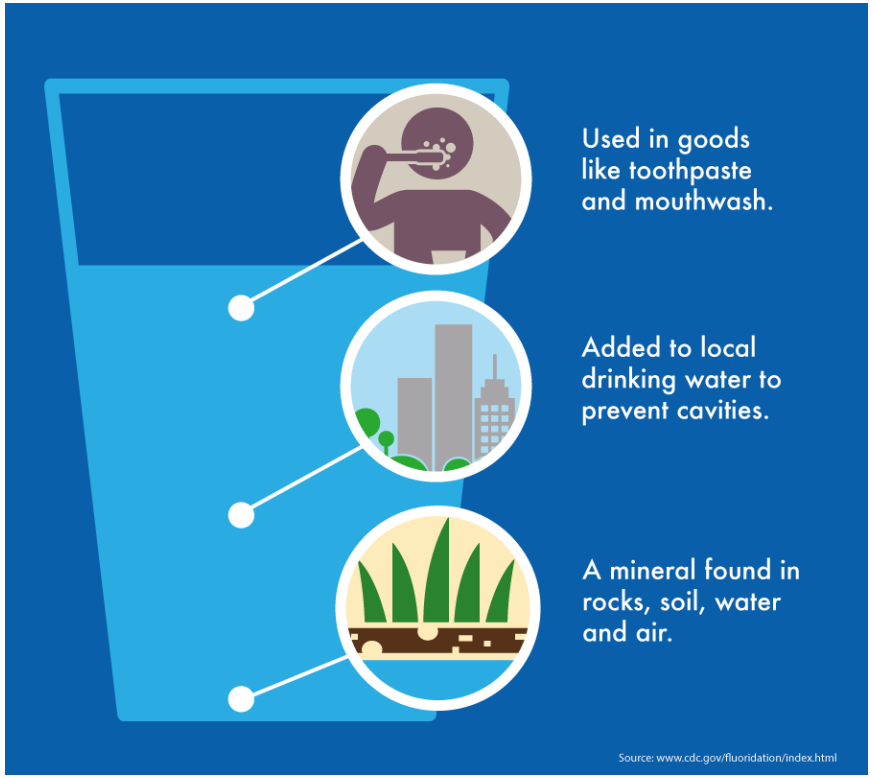
Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: Slideshare, https://www.slideshare.net/slideshow/lab5-determination-of-hardness-of-water/54958447#5

Role of Ammonia Buffer in EDTA titration

- A buffer solution is a solution that resists changes in its pH when small amounts of an acid or a base are added. Specifically, ammonia is often used to buffer the solution when the titration is conducted in slightly basic conditions, typically around pH 8 to 10, depending on the metal ion being titrated.
- In an EDTA titration, ammonia (or ammonia buffer) plays an important role in maintaining the pH of the solution within an optimal range (pH 10 in this case) for the complexation of metal ions by EDTA.

Fluoride

- Generally associated in nature with a few types of sedimentary or igneous rocks, fluoride is seldom found in appreciable quantities in surface waters and appears mainly in groundwater in only a few geographical regions.
- Fluoride is toxic to humans and other animals in large quantities, while small concentrations can be beneficial.
- Concentrations of approximately 1.0 mg/L in drinking water help to prevent dental cavities in children.
- During formation of permanent teeth, fluoride combines chemically with tooth enamel, resulting in harder, stronger teeth that are more resistant to decay.
- Fluoride is often added to drinking water supplies if sufficient quantities for good dental formation are not naturally present.



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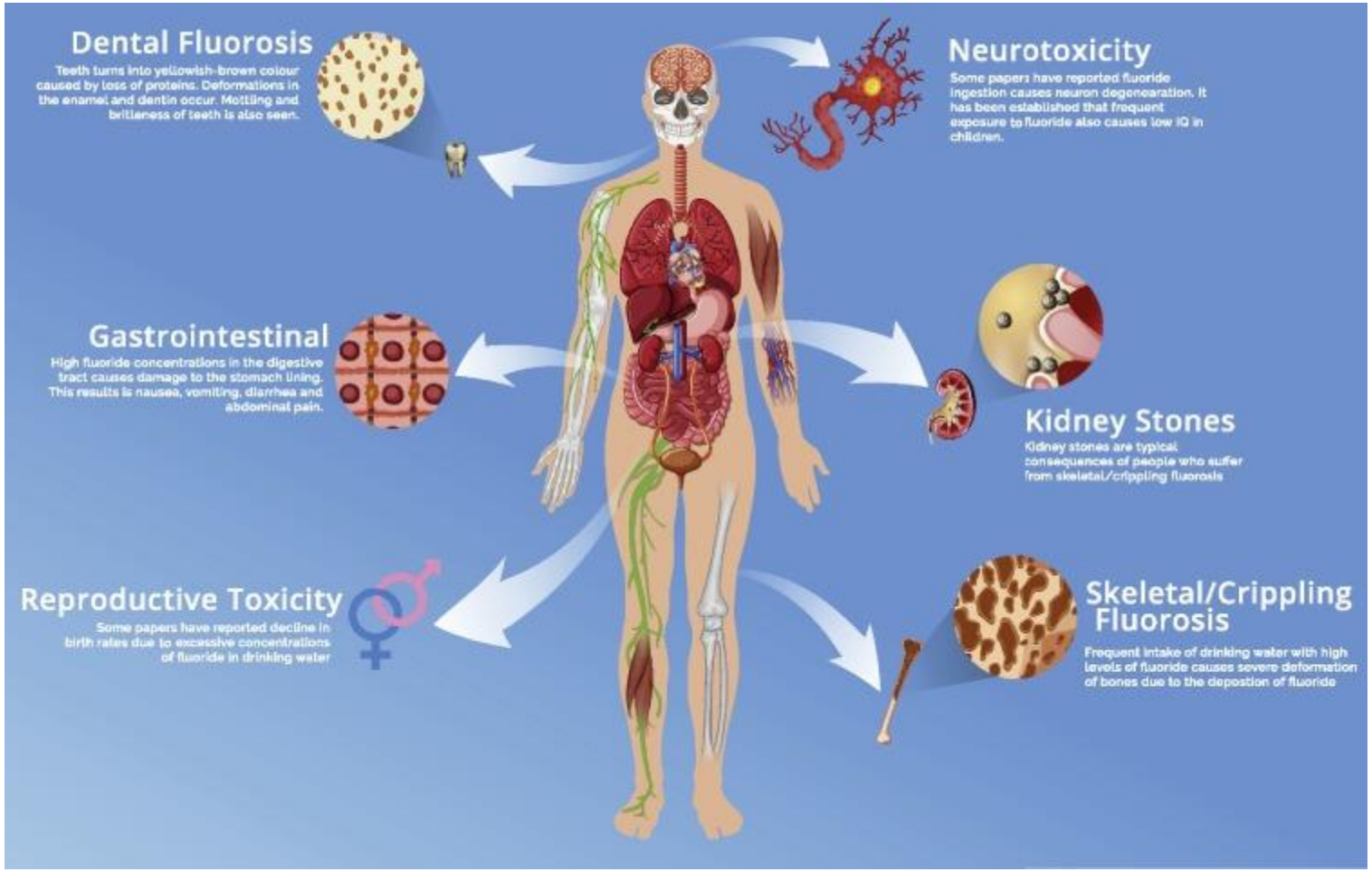
Fluoride

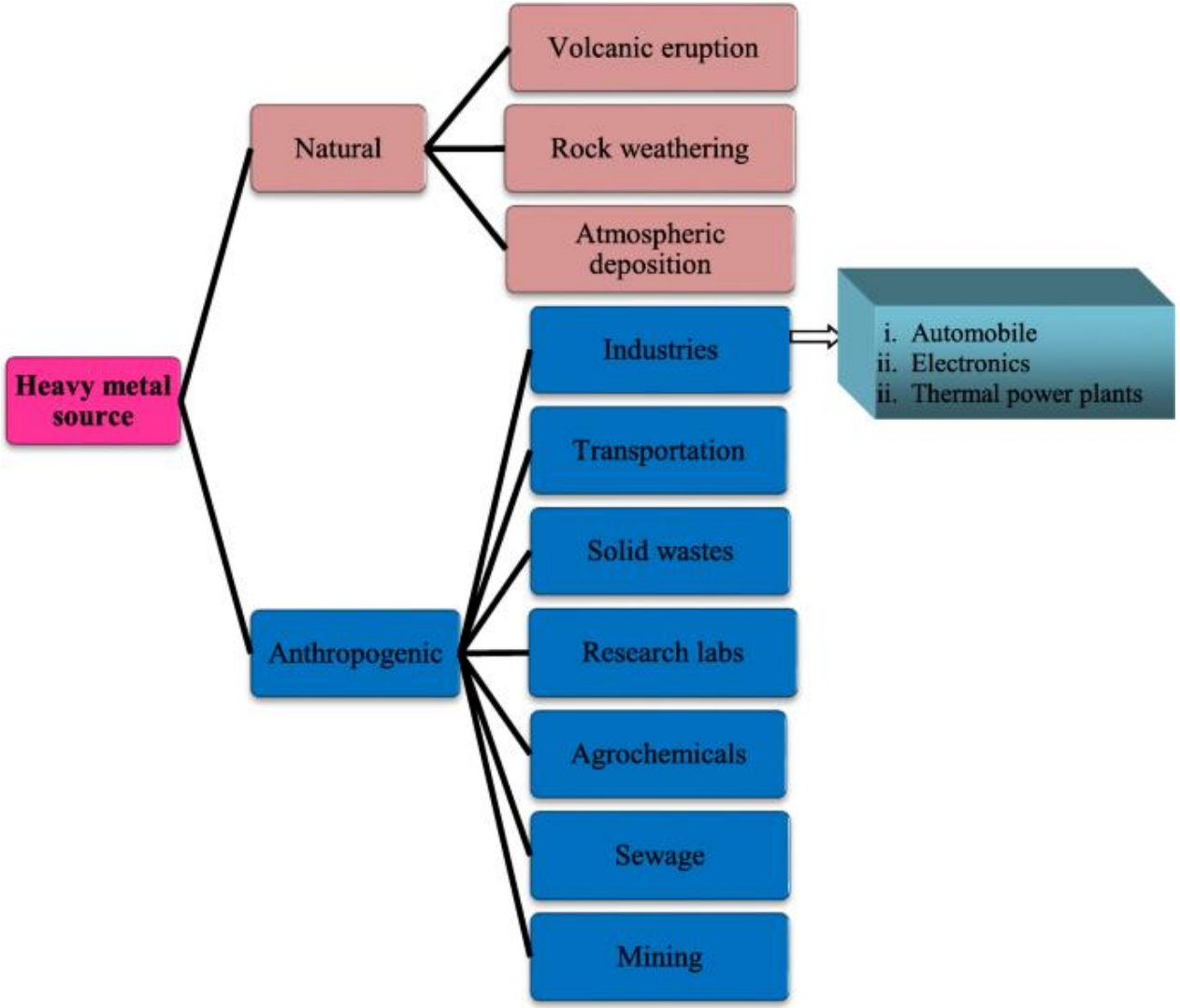
- Excessive intakes of fluoride can result in discoloration of teeth.
- Noticeable discoloration, called mottling, is relatively common when fluoride concentrations in drinking water exceed 2.0 mg/L, but is rare when concentrations are less than 1.5 mg/L.
- Excessive dosages of fluoride can also result in bone fluorosis and other skeletal abnormalities. Concentrations of less than 5 mg/L in drinking water are not likely.. to cause bone fluorosis or related problems.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: Kashyap, Shreyas J., Ravi Sankannavar, and G. M. Madhu. "Fluoride sources, toxicity and fluorosis management techniques–A brief review." Journal of Hazardous Materials Letters 2 (2021): 100033.

Metals

- All metals are soluble to some extent in water.
- While excessive amounts of any metal may present health hazards, *only those metals that are harmful in relatively small amounts are commonly labeled toxic; other metals fall into the nontoxic group.*
- Sources of metals in natural waters include dissolution from natural deposits and discharges of domestic, industrial, or agricultural wastewaters.
- Measurement of metals in water is usually made by *atomic absorption spectrophotometry*.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: Kanwar, Varinder Singh, et al. "Phytoremediation of toxic metals present in soil and water environment: a critical review." Environmental Science and Pollution Research 27 (2020): 44835-44860.

Metals

- **Toxic metals** that may be dissolved in water include arsenic, barium, cadmium, chromium, lead, mercury, and silver. Cumulative toxins such as arsenic, cadmium, lead, and mercury are particularly hazardous. These metals are concentrated by the food chain, thereby posing the greatest danger to organisms near the top of the chain.
- **Non-toxic metals:** Iron and manganese quite frequently occur together and present no health hazards at concentrations normally found in natural waters. Iron and manganese in very small quantities may cause color problems. Iron concentrations of 0.3 mg/L and manganese concentrations as low as 0.05 mg/L can cause color problems. Additionally, some bacteria use iron and manganese compounds for an energy source, and the resulting slime growth may produce taste and odor problems.

Element	EPA Limit (mgL^{-1})
Antimony	0.006
Arsenic	0.000
Beryllium	0.004
Chromium (total)	0.100
Cadmium	0.005
Cupper	1.300
Lead	0.000
Mercury	0.002
Selenium	0.050
Silver	0.100

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. EPA guidelines for drinking water, https://www.epa.gov/ground-water-and-drinking-water/national-primary-drinking-water-regulations

Numerical

Question No. 5: A 200 ml sample of water has an initial Ph of 10. 11 ml and 30 ml of 0.02N Sulphuric acid is required to titrate the sample to Ph 8.4 and 4.5 respectively. (a) Determine the species and the quantity of each species. (b) What is the total alkalinity of the sample?

Answer: (a)

1. Because the initial pH is 10, the initial pOH of the water is 4. A determination of the OH^- concentration can be made as follows

$$[OH^-] = \frac{10^{-4} mol\ OH-}{L} \times \frac{1\ equiv}{mol\ OH-} \times \frac{50{,}000\ mg\ CaCO3}{1\ equiv} = 5 \text{ mg/L as } CaCO_3.$$

2. 5 mL of 0.02N Sulphuric acid will be required to measure the OH in 1 liter of this water sample. However, the sample volume here is only 200 ml, so volume of acid needed

$$= \frac{5x200}{1000} = 1.0 \text{ mL}$$

3. If 1 mL of acid measure OH then 10 mL of the acid measure half of the carbonate (CO_3^{2-}), so 10 mL will be needed to measure the another half of the carbonate ion leaving 9 ml to measure or neutralize bicarbonate (HCO_3^-) ion.

$$[OH^-] = 5 mg/L\ as\ CaCO_3\ ,\ [CO_3^{2-}] = \frac{(10+10)\text{mg x } 1000\ m\ L/L}{200\ mL} = 100\ mg/L\ as\ CaCO_3$$

$$[HCO_3^-] = \frac{9 \text{ mg x}1000\ m\ L/L}{200\ mL} = 45\frac{mg}{L}\ as\ CaCO_3$$

(b) Total Alkalinity = 150 mg/L $as\ CaCO_3$

Source: 1. Howard S. Peavy et al, Environmental Engineering, McGraw Hill Education (2018).

Thank you

For any questions, please feel free to reach out to me.